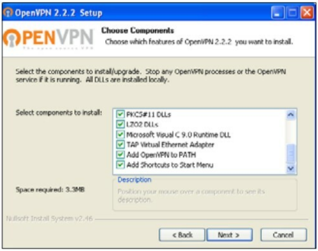

Figure 3: OpenVPN set-up options

install on IPCop 1.4.21, we need to tweak the installer. Edit the install file using any text editor. Find the *if* condition line with '1.4.15' (see Figure 1) and change it to '1.4.21'. Save and quit. Run *./install*, which adds an *OpenVPN* menu under *VPNs* in the IPCop Web GUI (HTTPS, port 445). Navigate here and the first step is to click the '*Generate Root/Host Certificates*' button. Fill up the required fields and generate the certificate.

Proceed to '*Roadwarrior Client status and control*' and add a client certificate for the desired host. Enter a complex password here; you will need it to establish the VPN connection. Now, the *VPNs* > *OpenVPN* page will show the completed configuration including root certificates, client certificates and buttons, and we can start/stop the OpenVPN server. Click *Start*. The OpenVPN server listens on UDP port 1194 for possible incoming connections from remote hosts. Figure 2 shows the running server's status page.

This completes Zerina's installation. Next, let's install and configure an OpenVPN client on a Windows desktop. Download the latest stable release installer for your OS (I used 32-bit Windows). Run the installer and complete the installation steps, including choosing all options, as in Figure 3.

From the IPCop Web GUI, download the earlier-generated OpenVPN client certificate's zip file (Figure 4), and extract its contents (two files, with extensions *.p12* and .ovpn) in the *C:\Program Files\OpenVPN\Config* folder.

Initiate a VPN connection by right-clicking the OpenVPN system-tray icon and choosing *Connect*. Enter the client certificate password to establish the VPN connection. You can verify the connection using a simple *ping* command to any office host's IP address.

View established VPN connections in the IPCop Web GUI—roadwarrior client status (Figure 5) is 'OPEN'. OpenVPN's connection statistics (Figure 6) show the last activity and data transfer details.

Real-life story

A company planned 75 VPN clients for its field personnel. After seeking a proposal from a commercial firewall vendor, they found that the client licensing cost was more than Rs 100,000. Here, installing OpenVPN-enabled IPCop was practically free! The

Figure 4: Download VPN certificates

Figure 5: OpenVPN client connections

Figure 6: OpenVPN connection statistics

IPCop was installed as a VPN gateway, BlockOutgoing Traffic was used to allow only OpenVPN traffic.

Tips for OpenVPN installation

Client side:

- On client systems, configure desktop firewalls to allow OpenVPN traffic.
- Run the OpenVPN installer, as well as start OpenVPN, as the administrator—for Windows Vista and later versions.

Server side:

- If your VPN connection drops intermittently, visit *VPNs* > *OpenVPN* in the IPCop Web GUI. Stop the VPN server, configure *Advanced Server Options* with *keep-alive* and *ping/ping-restart* with 10 and 60 seconds respectively. Save the options and start the OpenVPN server again.
- Typically, VPN requires static IP address on server side. IPCop supports DynDNS service, which can be leveraged to setup VPN server on dynamic IP addresses. END

Links

[1] Zerina download: *http://www.openvpn.eu/uploads/media/ZERINA-0.9.7a14-Installer.tar.gz*
[2] Download page for OpenVPN 2.2.2: *http://openvpn.net/index.php/open-source/downloads.html*

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